

Benchmark #5

300keV Xe on UO2

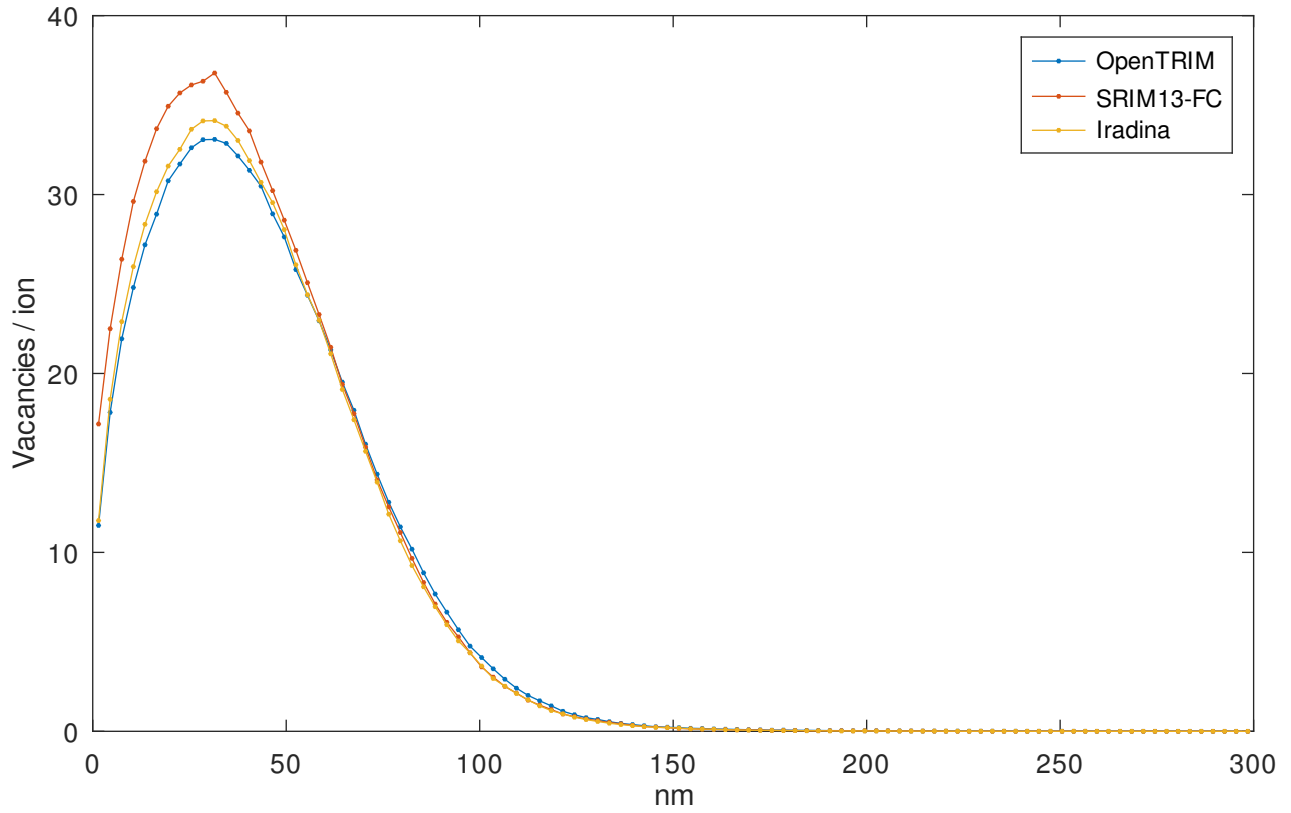
Ion energy E0 = 300000 eV

Target depth = 300 nm

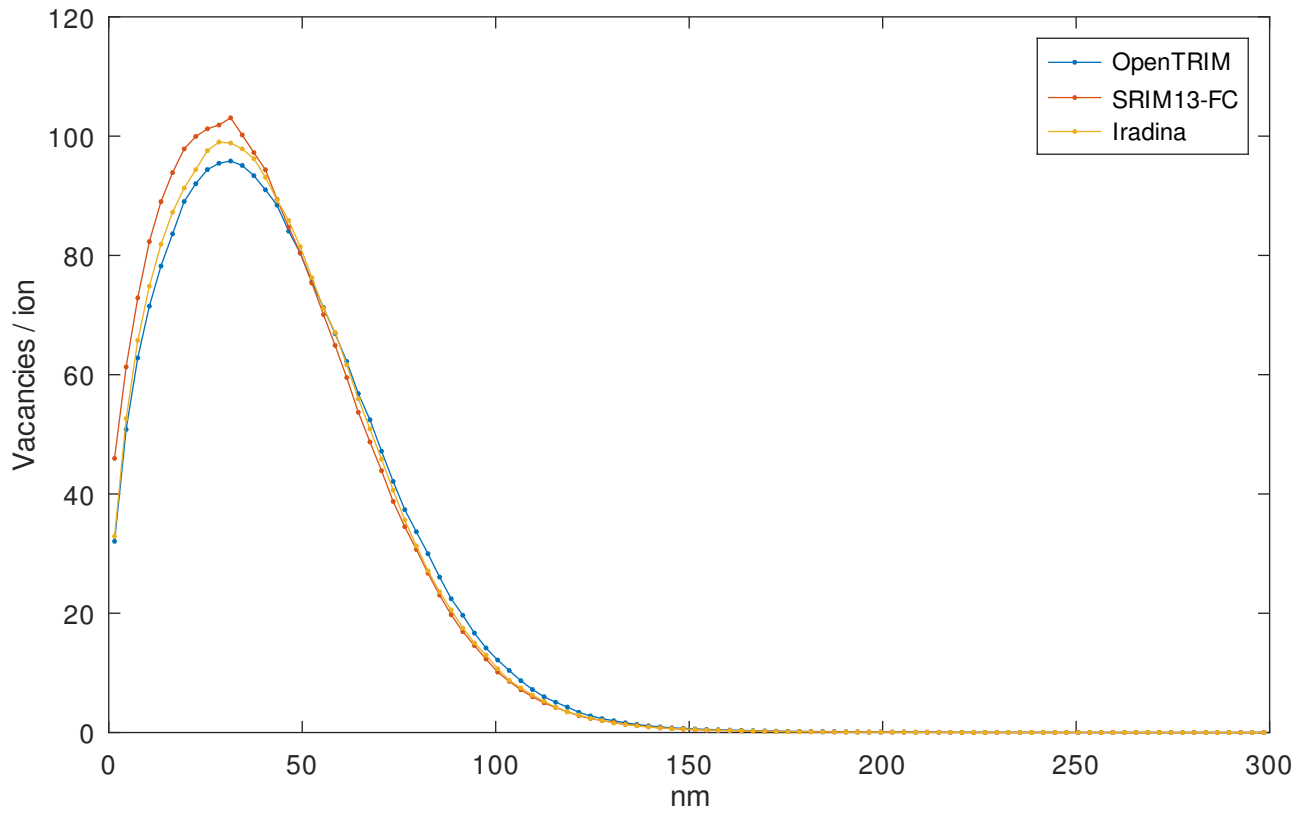
Summary Table

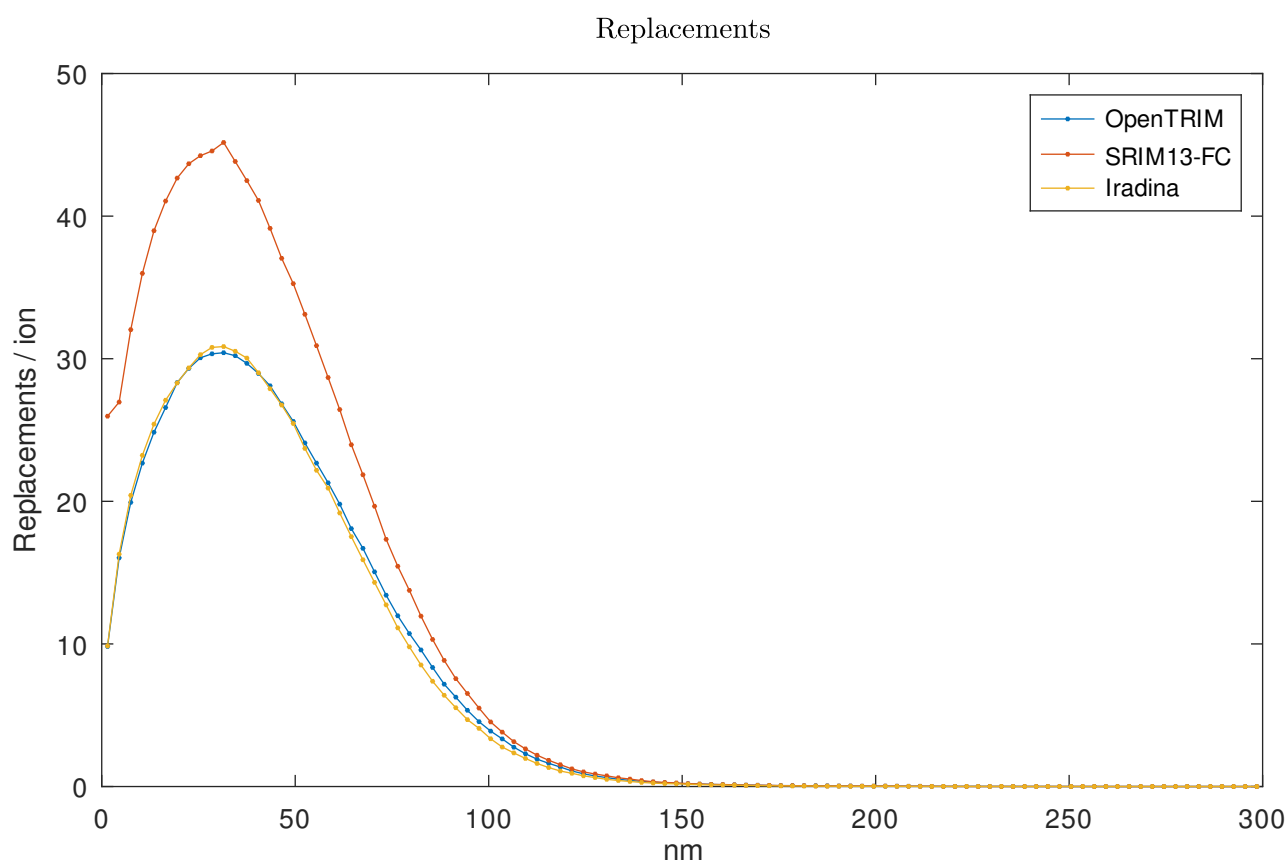
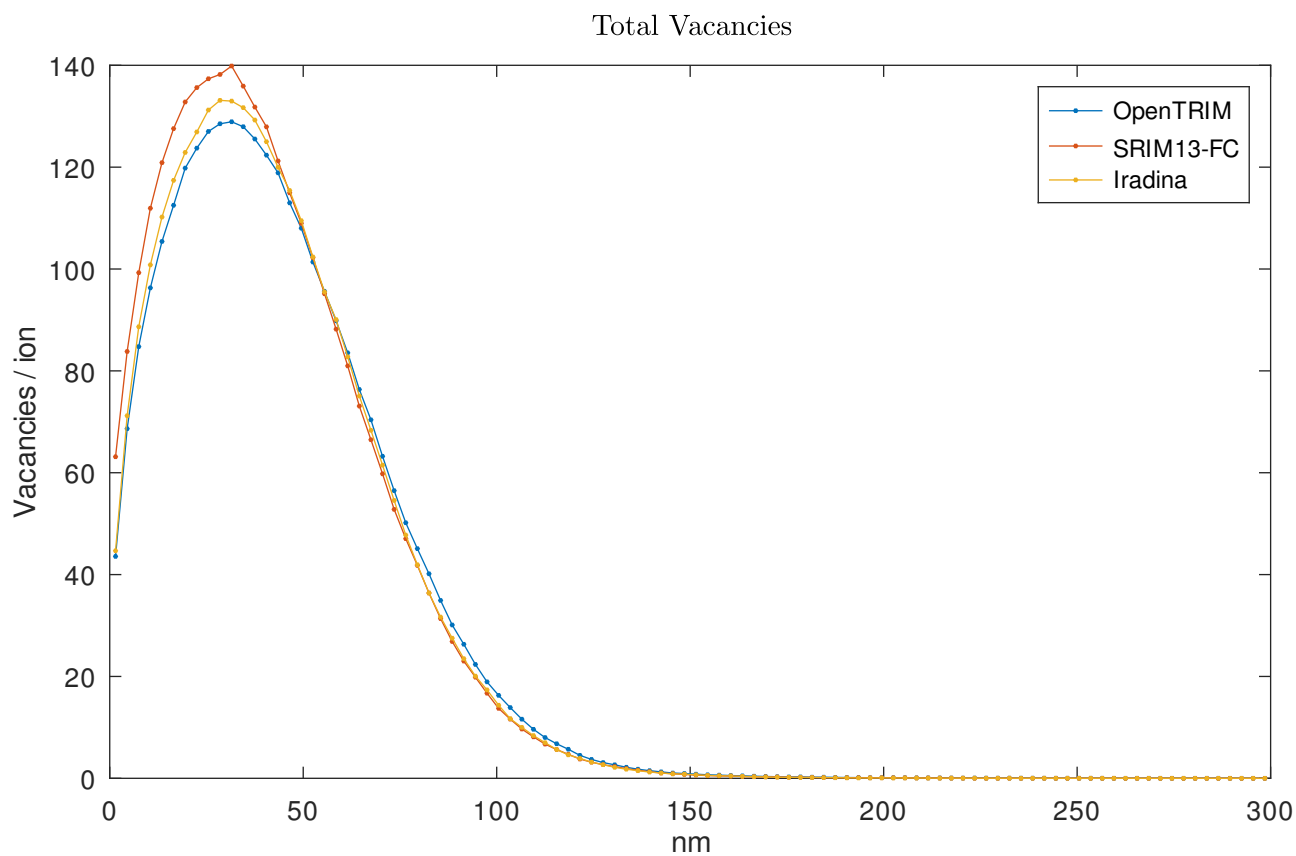
Quantity	OpenTRIM	SRIM13-FC	Iradina
V(U)	732	786	735
V(O)	2.13e+03	2.19e+03	2.14e+03
V(tot)	2.86e+03	2.98e+03	2.87e+03
R(tot)	677	970	666
I(Xe)	0.985	0.978	0.979
$\Delta E_e(\text{Xe})/E_0$	0.102	0.1	0
$\Delta E_e(r)/E_0$	0.337	0.325	0
$\Delta E_e(\text{tot})/E_0$	0.439	0.425	0.43
$\Delta E_{ph}(\text{Xe})/E_0$	0.00415	0.00366	0
$\Delta E_{ph}(r)/E_0$	0.544	0.518	0
$\Delta E_{ph}(\text{tot})/E_0$	0.548	0.522	0.555
$1 - (\Delta E_e + \Delta E_{ph})/E_0$	0.0134	0.0527	0.0151

Vacancies of U in Uranium oxide

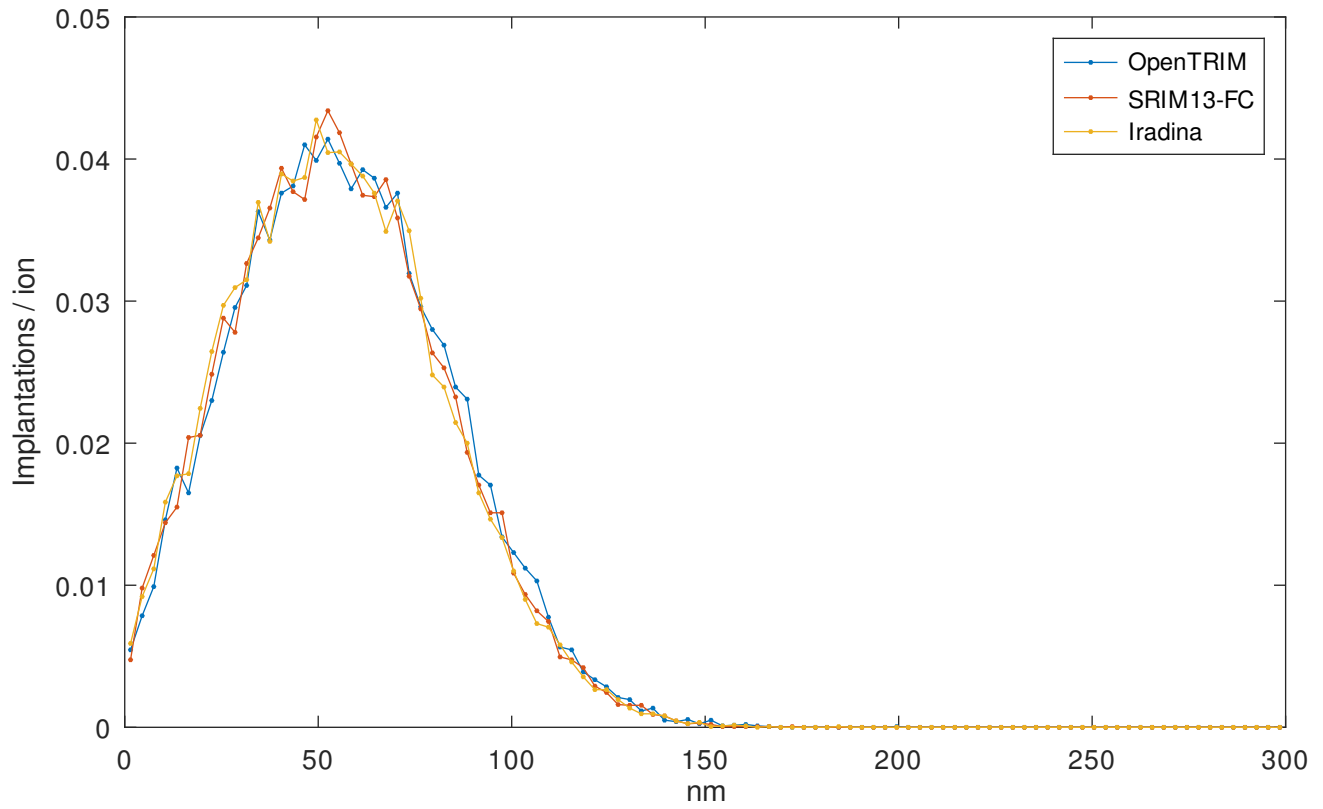


Vacancies of O in Uranium oxide

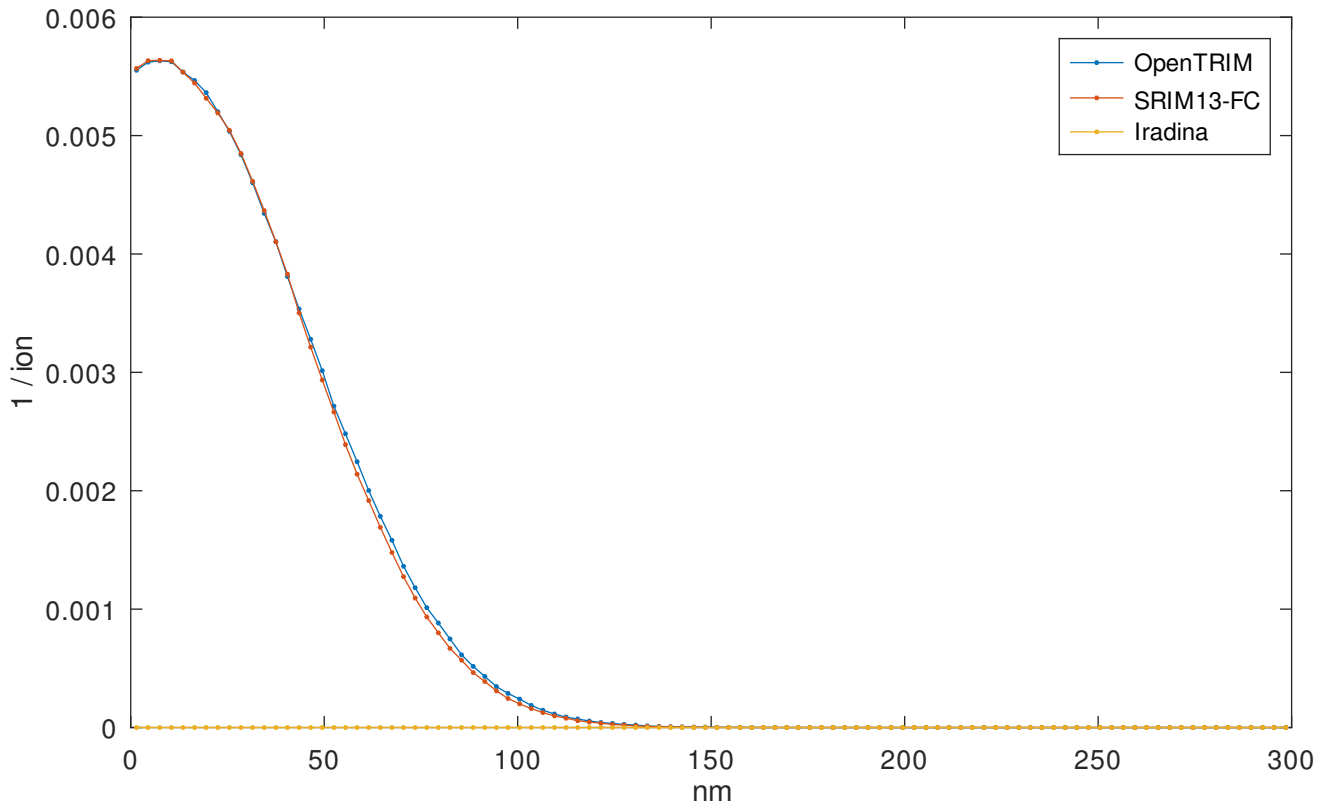




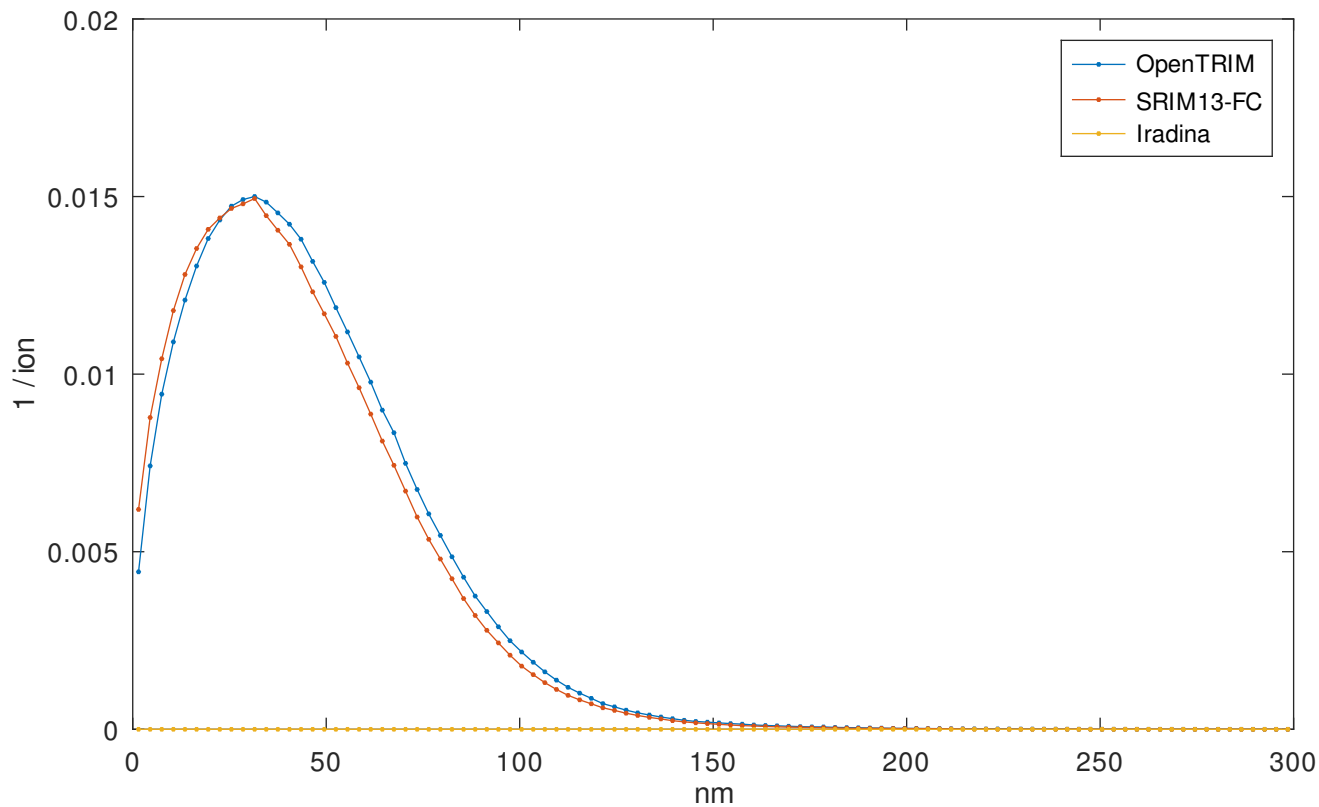
Implanted Xe ion



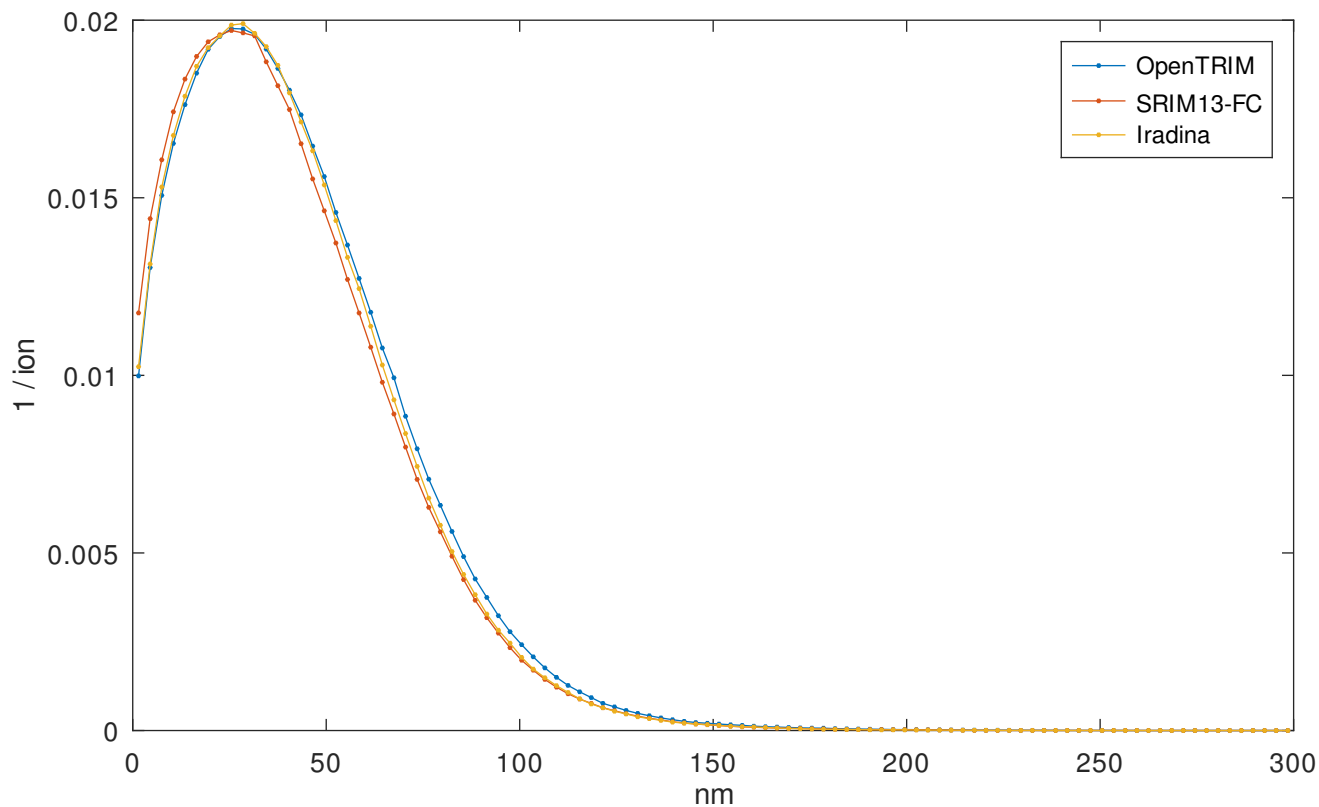
Electronic energy loss fraction $\Delta E_e/E_0$ by Xe ion



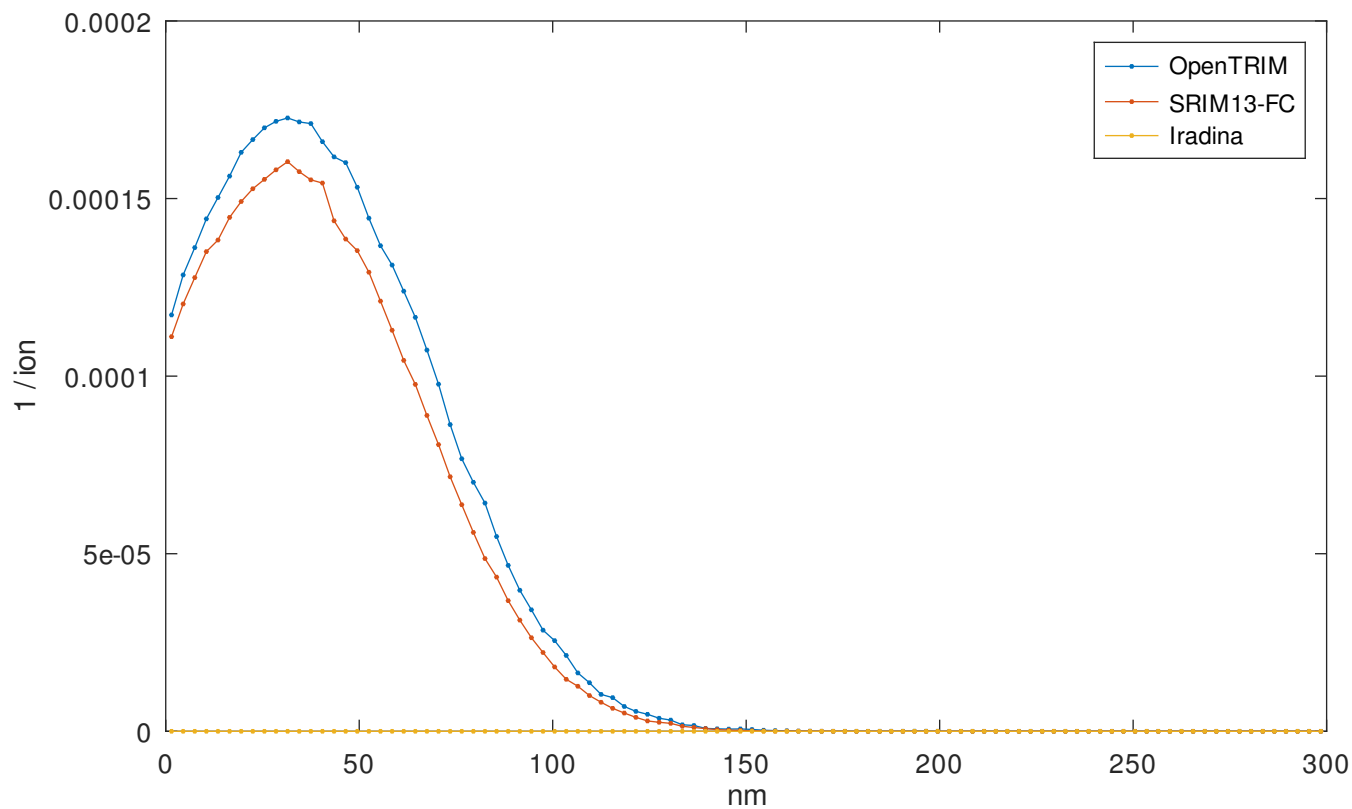
Electronic energy loss fraction $\Delta E_e/E_0$ by recoils



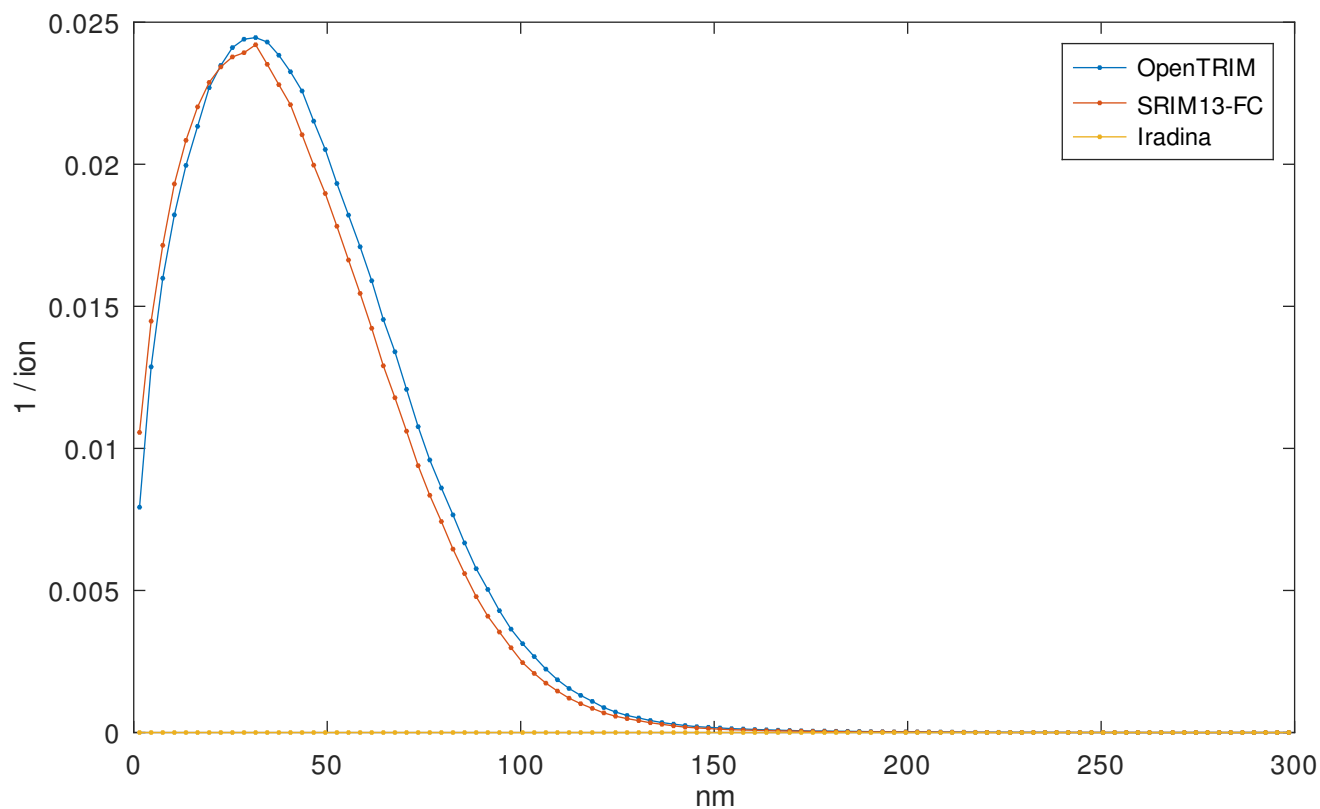
Total Electronic energy loss fraction $\Delta E_e/E_0$



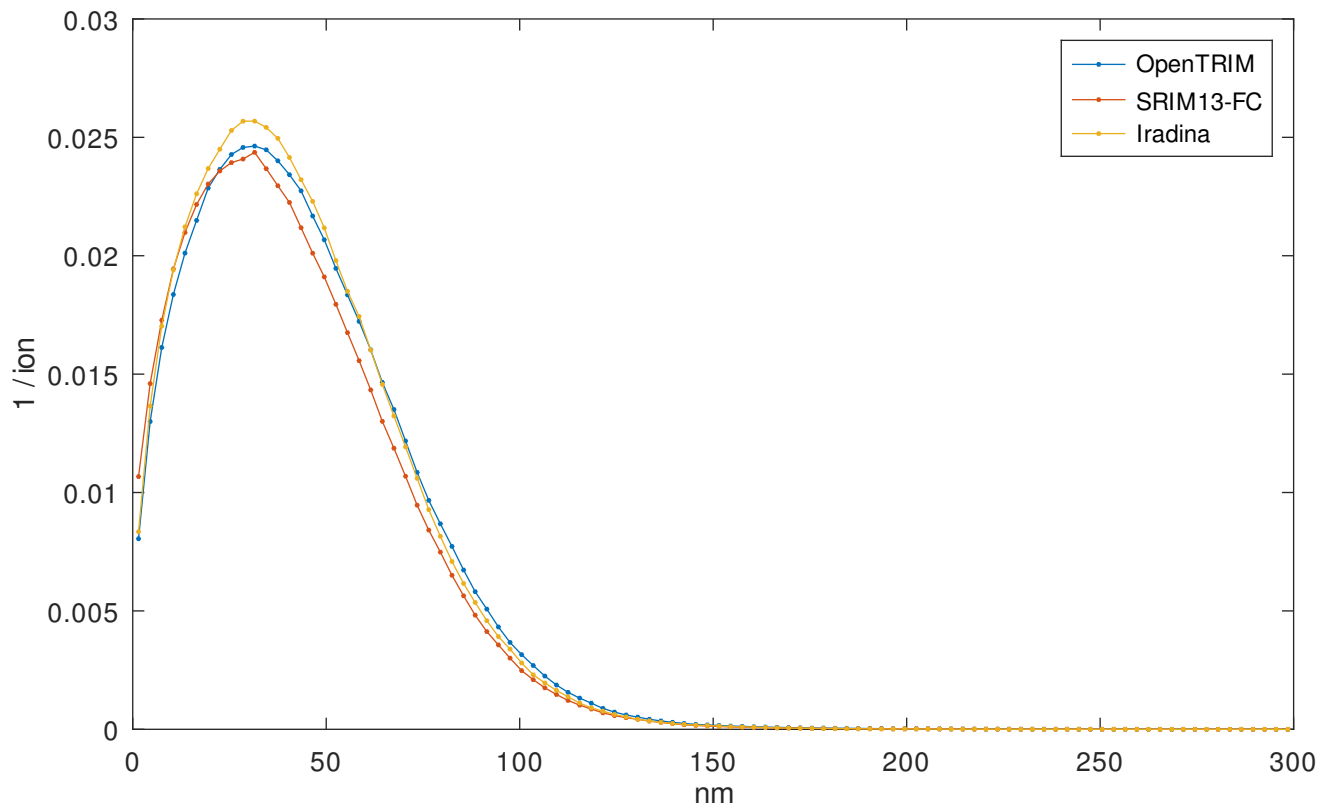
Phonon energy fraction $\Delta E_{Ph}/E_0$ by Xe ion



Phonon energy fraction $\Delta E_{Ph}/E_0$ by recoils



Total Phonon energy fraction $\Delta E_{Ph}/E_0$



Total fractional energy deposition $(\Delta E_e + \Delta E_{Ph})/E_0$

